Software Requirements Specification (SRS)

Project: Project Management System (PMS)
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Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Nidhi Raghuraman, Padarthi Seshank, , Prabhanjana P Hegde, Prajwal R	SRS with diagrams, security requirements and udated RTM	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose: This document specifies the requirements for a web-based Project Management System for managing projects, users, and tasks.

1.2 Scope: The system allows users to create projects, manage team members, assign tasks, track task status, and monitor project progress.

1.3 Audience: Developers, testers, project members, and evaluators.

1.4 Definitions: PMS – Project Management System UI – User Interface DB – Database

2. Overall description

2.1 Product Perspective The system is a web application consisting of a frontend, backend, and PostgreSQL database.

2.2 Major Functions

- User registration and login
- Create and manage projects
- Add project members
- Create and assign tasks
- Update task status
- Set task priority and deadlines
- View project progress

2.3 User Roles

- Administrator - Creates and manages projects
- Project Manager - Creates and manages tasks
- Task Manager - Manages users assigned to tasks

2.4 Operating Environment Web browser, Windows/Linux/macOS, Node.js server, and PostgreSQL database.

2.5 Constraints The system requires a network connection and a PostgreSQL database.

3. External interface requirements

3.1 User Interface Web-based dashboard with pages for projects, tasks, users, and project progress.

3.2 Software Interfaces

- Next.js application

PostgreSQL database

3.3 Communication Client-server communication through HTTP/HTTPS.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow PMS-F-###.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref
PMS-F-001	The system shall allow users to register and log in.	Functional	High	User	User can register with valid details and log in with correct credentials.
PMS-F-002	The system shall allow users to manage their profiles.	Functional	High	User	User can view and update their profile information.
PMS-F-003	The system shall allow Administrator to create Projects.	Functional	Medium	Administrator	Administrator can create a project with valid project details.
PMS-F-004	The system shall allow users to view projects they belong to.	Functional	High	User	User can view only projects they belong to.
PMS-F-005	The system shall allow project manager to create tasks.	Functional	High	Project Manager	Project Manager can create a task within a project.

PMS-F-006	The system shall allow task managers to assign tasks to users	Functional	High	Task Manager	A task can be assigned to a valid user .
PMS-F-007	The system shall allow users to view tasks assigned to them.	Functional	High	User	User can view the tasks assigned to them.
PMS-F-008	The system shall allow users to update task status.	Functional	High	User	User can change the status of an assigned task.
PMS-F-009	The system shall allow tasks to have priorities and deadlines.	Functional	Medium	User	Task can have a priority and deadline, and these values can be viewed.
PMS-F-010	The system shall display project progress.	Functional	Medium	User	System displays project progress correctly based on completed tasks
PMS-F-011	The system shall allow administrators to manage project details.	Functional	Medium	Administrator	Administrator can view and update valid project details.
PMS-F-012	The system shall allow administrators to add project members.	Functional	High	Administrator	Administrator can add a valid user to a project.

PMS-F-013	The system shall allow project managers to set task priority and deadlines.	Functional	Medium	Project Manager	Project Manager can set or update a task priority and deadline.
PMS-F-014	The system shall allow users to view the status of tasks in their projects.	Functional	Medium	User	User can view the current status of tasks they are authorized to access.
PMS-F-015	The system shall allow users to view project progress based on task completion.	Functional	Medium	User	User can view project progress calculated from completed tasks.

5. Non-functional requirements (detailed)
NFRs below are measurable and tied to test plans. IDs PMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
PMS-NF-001	The system shall respond to normal user requests within 3 seconds.	Performance	High	Normal user requests receive a response within 3 seconds under normal load.
PMS-NF-002	The system shall prevent unauthorized access to projects and tasks.	Security	High	Users cannot access projects or tasks they are not authorized to access.

PMS-NF-003	The system shall maintain consistent project and task data.	Reliability	High	Project and task data remains correct and consistent after creation or modification.
PMS-NF-004	The system shall support multiple users simultaneously.	Scalability	Medium	Multiple users can access and use the system simultaneously without significant performance issues.
PMS-NF-005	The interface shall be simple and easy to navigate.	Usability	Medium	Users can navigate the main system functions without unnecessary complexity or confusion.

5.1 Security

5.1.1 Security Objectives

Objective ID	Security Objective	Acceptance Criteria
SO-001	Protect project and task data from unauthorized access.	Only authorized users can access project and task data.
SO-002	Protect user authentication and profile information from unauthorized access or modification.	Only authenticated and authorized users can access or modify protected user information.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref	Status
PMS-SR-001	The system shall restrict access to projects and tasks based on user authorization.	Security	High	Users cannot access projects or tasks they are not authorized to access. Test: TC-SEC-01	N
PMS-SR-002	The system shall require user authentication before allowing access to protected system functions.	Security	High	Unauthenticated users cannot access protected project and task functions. Test: TC-SEC-02	N
PMS-SR-003	The system shall allow users to view and modify only their own profile information.	Security	High	A user cannot modify another user's profile. Test: TC-SEC-03	N
PMS-SR-004	The system shall enforce role-based permissions for Administrator, Project Manager, and Task Manager functions.	Security	High	Users can perform only functions permitted by their assigned role. Test: TC-SEC-04	N
PMS-SR-005	The system shall protect project and task data during client-server communication using HTTP/ HTTPS as applicable.	Security	High	Protected data is transmitted through the configured client-server communication channel. Test: TC-SEC-05	N

SO-001: Protect project and task data from unauthorized access.

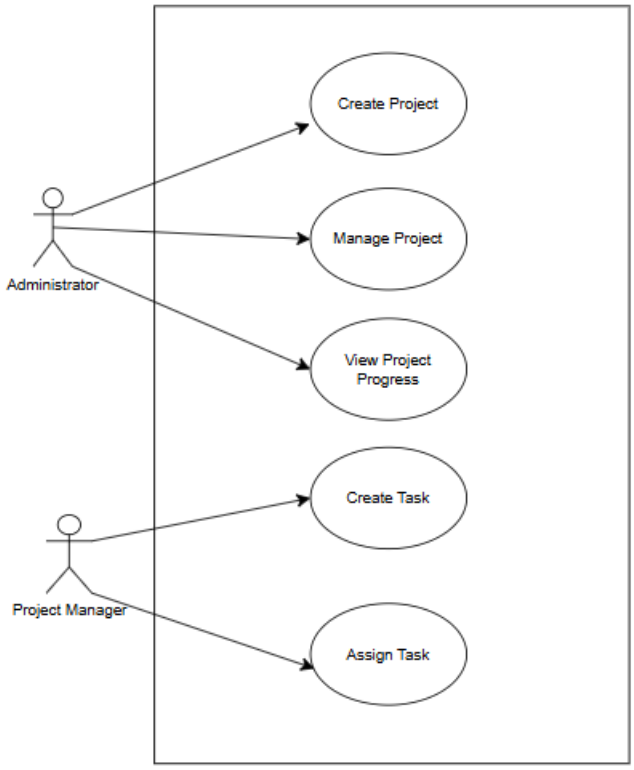
SO-002: Protect user authentication and profile information from unauthorized access or modification.

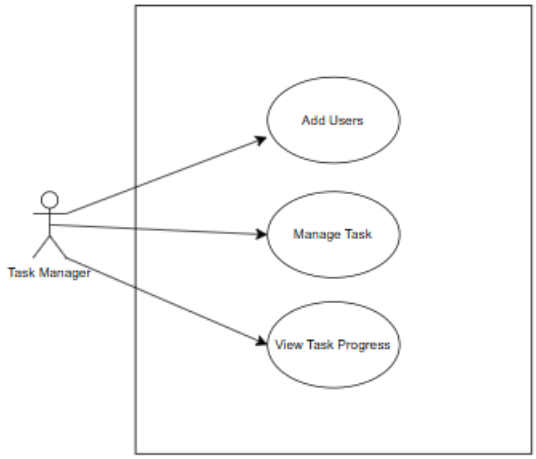
6. Quality attributes & Acceptance tests

Users can successfully log in.
Projects can be created and managed.
Tasks can be created and assigned.
Task status can be updated.
Project progress is displayed correctly.
Unauthorized users cannot access restricted data.
All high-priority requirements are implemented and tested

Security objectives and all five security requirements are included and mapped in the RTM.

7. System models and diagrams





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Module	Test case(s)	Status (N/P/A)
PMS-F-001	User login	Authentication	TC-01	N
PMS-F-003	Create project	Project Management	TC-02	N
PMS-F-005	Create task	Task Management	TC-03	N
PMS-F-006	Assign user to Task	Task Management	TC-04	
PMS-F-008	Update task status	Task Management	TC-05	N
PMS-NF-001	Response time	Web Application	TC-06	N